

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA0711

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 40%

Code of Conduct:

I Nagarajan Subhash / 192312489 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

N. Subhash
Signature of the student:

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks	Q5 10marks
Problem Understanding	20%	1.5	1.5	1.5	1.5	1.5
Method Selection	20%	1.5	1.5	1.5	1.5	1.5
Solution Process	25%	1.5	1.5	1.5	1.5	1.5
Accuracy of Calculation	20%	2.0	2.0	2.0	2.0	2.0
Interpretation of Results	15%	1/2	1/2	1/2	1/2	1/2

Total Marks 30 /40

[Signature]
Signature of the Faculty

Annexure A

ANALYTICAL PROBLEM SOLVING

1. A university network is assigned the IP address block 192.168.20.0/24 for its internal communication. The network administrator has decided to divide this network into subnets for three different labs using fixed subnet masks. Lab A is assigned a /26 subnet, Lab B is assigned a /27 subnet, and Lab C is assigned a /28 subnet. Based on the given subnet masks, apply subnetting concepts to determine the network address of each lab. Further, identify the valid range of host IP addresses and the broadcast address for each subnet. Finally, calculate the total number of usable host IP addresses available in each lab's subnet.
2. A network administrator has configured two systems with the following details: System 1 has IP address 192.168.1.130 with subnet mask 255.255.255.192, and System 2 has IP address 192.168.1.190 with subnet mask 255.255.255.192. Using the given configuration, apply subnetting techniques to determine the subnet range, network address, and broadcast address for each system. Identify the valid host IP range for both subnets based on the given subnet mask. Calculate whether each IP address falls within its respective subnet range. Also, determine the number of usable host addresses available in each subnet.
3. An organization has been allocated the IP block **172.16.0.0/16** for its internal network. The network administrator has already decided the subnet masks for each department as follows: HR → /25, Sales → /26, IT → /27, and Admin → /28. Using the given subnet masks, apply subnetting techniques to assign suitable subnet addresses for each department. Determine the network address, valid host IP range, and broadcast address for each subnet. Based on the allocation, calculate the number of usable host addresses in each department.

Finally, identify the remaining unused IP address range within the given network block.

4. A company uses the IP address **172.16.10.0/24** and has implemented subnetting using a **/27 subnet mask** for its internal departments. Using the given subnet mask, apply subnetting techniques to calculate the total number of subnets created and the number of usable hosts per subnet. Determine the subnet to which the IP address **172.16.10.78** belongs. Identify the network address and broadcast address for that subnet. Also, determine whether the IP address **172.16.10.95** falls within the same subnet range based on the given configuration.

5. A network uses the Internet Protocol version 6 address **2001:0db8:0000:0000:0000:ff00:0042:8329**. Apply IPv6 addressing rules to compress the given address into its shortest form and then expand the compressed address back to its full notation. Using a prefix length of **/64**, determine the network prefix of the given address. Also, calculate the total number of possible host addresses available within this network based on the given prefix length.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method

Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	N calo
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	N interpr

Subnetting of 192.168.20.0/24

LABA (/26)

- Subnet Mask: 255.255.255.192
- Network Address: 192.168.20.0
- Valid Host Range: 192.168.20.1 - 192.168.20.62
- Broadcast address: 192.168.20.63
- Usable Hosts: 62

~~Lab B (/27)~~

- Subnet.

Formula

- Number of Host Bits = 32 - prefix length
- Total Addresses = $2^{\text{host bits}}$
- Usable Hosts = $2^{\text{host bits}} - 2$

Block size = 256 - last Octet of Subnet Mask

Prefix	Subnet Mask	Block size
126	255.255.255.192	64
127	255.255.255.224	32
128	255.255.255.240	16

Lab B (/27)

Subnet Mask: 255.255.255.224

Block size: 32

Next Available network: 192.168.20.64

Total ... 192.168.20.63

Last host: 192.168.209.5
Broadcast Address: 192.168.209.5

$$= 2^5 - 2$$

$$= 32 - 2$$

$$= 30 \text{ hosts}$$

Lab C (/28)

Subnet Mask: 255.255.255.240

Block size: 16

Next Available Network: 192.168.20.96

First Host: 192.168.20.110

Usable Hosts: $2^{(32-28)} - 2$

$$= 2^4 - 2$$

$$= 16 - 2$$

$$= 14 \text{ hosts}$$

Accuracy of calculation

• Lab A: $2^6 - 2 = 62$ usable hosts

• Lab B: $2^5 - 2 = 30$ usable hosts

• Lab C: $2^4 - 2 = 14$ usable hosts

Interpretation of results

The subnetting process successfully divides the original 192.168.20.0/24 network into three separate subnets. Lab A receives the most largest subnet because it requires the greatest no. of hosts, in the university environment.

Problem Statement

- A network administrator has configured two systems with the following details.
- System 1: IP Address = 192.168.1.130 Subnet Mask = 255.255.255.192
- System 2: IP Address = 192.168.1.190, Subnet Mask = 255.255.255.192 (/26)

Method Selection

Formula

- Host bits = 32 - prefix length
- Usable Hosts = $2^{(\text{Host bits})} - 2$
- Block size = 256 - 192 = 64

Subnet mask:

255.255.255.192 (/26)

Each subnet contains.

- Total IP Address = 64
- Usable Host Address = 62

Subnet Address Range

192.168.1.0/26	192.168.1.0 - 192.168.1.63
192.168.1.64/26	192.168.1.64 - 192.168.1.127
192.168.1.128/26	192.168.1.128 - 192.168.1.191
192.168.1.192/26	192.168.1.192 - 192.168.1.255

System 1

IP Address: 192.168.1.130

• Network Address: 192.168.1.128

• Last host: 192.168.1.190

• Broadcast Address: 192.168.1.191

Therefore, 192.168.1.130 is a valid host address

System 2

IP Address: 192.168.1.190

- Network Address: 192.168.1.128

• first host: 192.168.1.129

• last host: 192.168.1.190

• Broadcast Address: 192.168.1.191

Accuracy of calculation

$$\text{Host bits} = 32 - 26 = 6$$

$$\text{Usable hosts} = 2^6 - 2 = 64 - 2 = 62$$

$$\text{Network Address} = 192.168.1.128$$

$$\text{Broadcast Address} = 192.168.1.191$$

All subnet calculation are correct.

Interpretation of results

The analysis shows that both 192.168.1.130 and 192.168.1.190 belong to the same 192.168.1.128/26 subnet. Since both addresses are within the valid host range, they can communicate directly without requiring a router, while reducing unnecessary broadcast traffic.

Problem understanding

An organization has been allocated the IP block 172.16.0.0/16 for its internal network. The network administrator has assigned the following subnet masks.

- HR $\rightarrow /25$
- Sales $\rightarrow /26$
- IT $\rightarrow /27$
- Admin $\rightarrow /28$

Formula

- Host bits = $32 - \text{prefix length}$
- Total Address = $2^{\text{Host bits}}$
- Usable hosts = $2^{\text{Host bits}} - 2$
- block size = 256 last octet of subnet mask.

Solution process

HR Department (/25)

- Subnet mask: 255.255.255.128
- Network Address: 172.16.0.0
- first host: 172.16.0.1
- last host: 172.16.0.126
- Broadcast Address: 172.16.0.127
- Usable host: $2^7 - 2 = 126$

Sales Department (126)

- Subnet Mask: 255.255.255.192
- Network Address: 192.16.0.128
- first host: 192.16.0.129
- usable hosts: $2^6 - 2 = 62$

IT Department (128)

- Subnet Mask: 255.255.255.240
- Network Address: 192.16.0.24
- first host: 192.16.0.25
- usable host: $2^4 - 2 = 14$

Remaining unused IP Address range

- Starting Address: 192.16.0.240
- ending Address: 192.16.255.255

Accuracy of calculation

- HR (125): $2^7 - 2 = 126$ usable hosts
- Sales (126): $2^6 - 2 = 62$ usable hosts
- IT (127): $2^5 - 2 = 30$ usable hosts
- Admin (128): $2^4 - 2 = 14$ usable hosts

Interpretation of results

The network has been efficiently divided into four subnets according to the requirement of the HR, Sales, IT and Admin departments.

problem understanding

The company has a class C network 172.16.10.0/24.

To improve network organization, it is divided into smaller using /27 subnet mask. The task is to calculate the number of subnets, determine the number of usable hosts in each subnet.

Method selection

formula

- Borrowed bits = New prefix - Original prefix
- No of subnets = $2^{(\text{Borrowed bits})}$
- Host bits = $32 - \text{prefix length}$
- usable hosts = $2^{(\text{Host bits})} - 2$
- Block size = $256 - \text{last octet of subnet mask}$

Given:-

- Original network = 172.16.10.0/24
- New prefix = /27
- subnet mask = 255.255.255.224
- Block size = $256 - 224 = 32$

Solution process

$$\text{Borrowed bits} = 27 - 24 = 3$$

$$\text{No. of subnets} = 2^3 = 8$$

Step 2 :- calculate usable hosts

$$\text{Host bits} = 32 - 27 = 5$$

$$\text{usable Hosts} = 2^5 - 2 = 32 - 2 = 30$$

Step 3 :-

Therefore

- Network Address: 172.16.10.64
- valid host range: 172.16.10.65 - 172.16.10.94
- Broadcast Address: 172.16.10.95

4. Accuracy of calculation

- Borrowed bits = 3
- No. of subnets = $2^3 = 8$
- usable hosts = $2^5 - 2 = 30$
- Broadcast Address = 172.16.10.95

5. interpretation of results

Subnetting the 172.16.10.0/24 network into eight 27 subnets provides 30 usable host addresses in each subnet, making the network suitable for different departments.

problem understanding

The given address is an IPv6 address, which consists of 128 bits represented as eight groups of four hexadecimal digits. The task is to apply the number of possible host (interface) addresses in the network.

Method selection

Rules for IPv6 compression

1. Remove leading zeros from each hexadecimal block.
2. Replace one continuous sequence of zero blocks with only one in address.

Formulae

- Total IPv6 Address length = 128 bits
- prefix length = 64 bits
- Host bits = $128 - 64 = 64$
- No. of interface addresses = $2^{(\text{host bits})}$

Solution process

Step 1:

2001:0db8:0000:0000:0000:f00:0002:839

Calculate the NO. OF interface Address

$$\text{Host bits} = 128 - 64 = 64$$

NO. OF interface addresses

$$= 2^{64}$$

$$= 18,446,744,073,709,551,616$$

= This is approximately 18.4 Quintillion interface

Accuracy of calculation

- IPv6 address compressed correctly according to IPv6 rules
- expand address matches the original IPv6 address
- Network prefix correctly identified as.
- Host bits calculated as 64.

All calculation and IPv6 addressing operations were correct.

Interpretation of Results

The IPv6 address has been correctly compressed and expanded using standard IPv6 notation rules. With a 64 prefix, the first 64 bits are used to interface addressing internet of Things (IoT) applications.